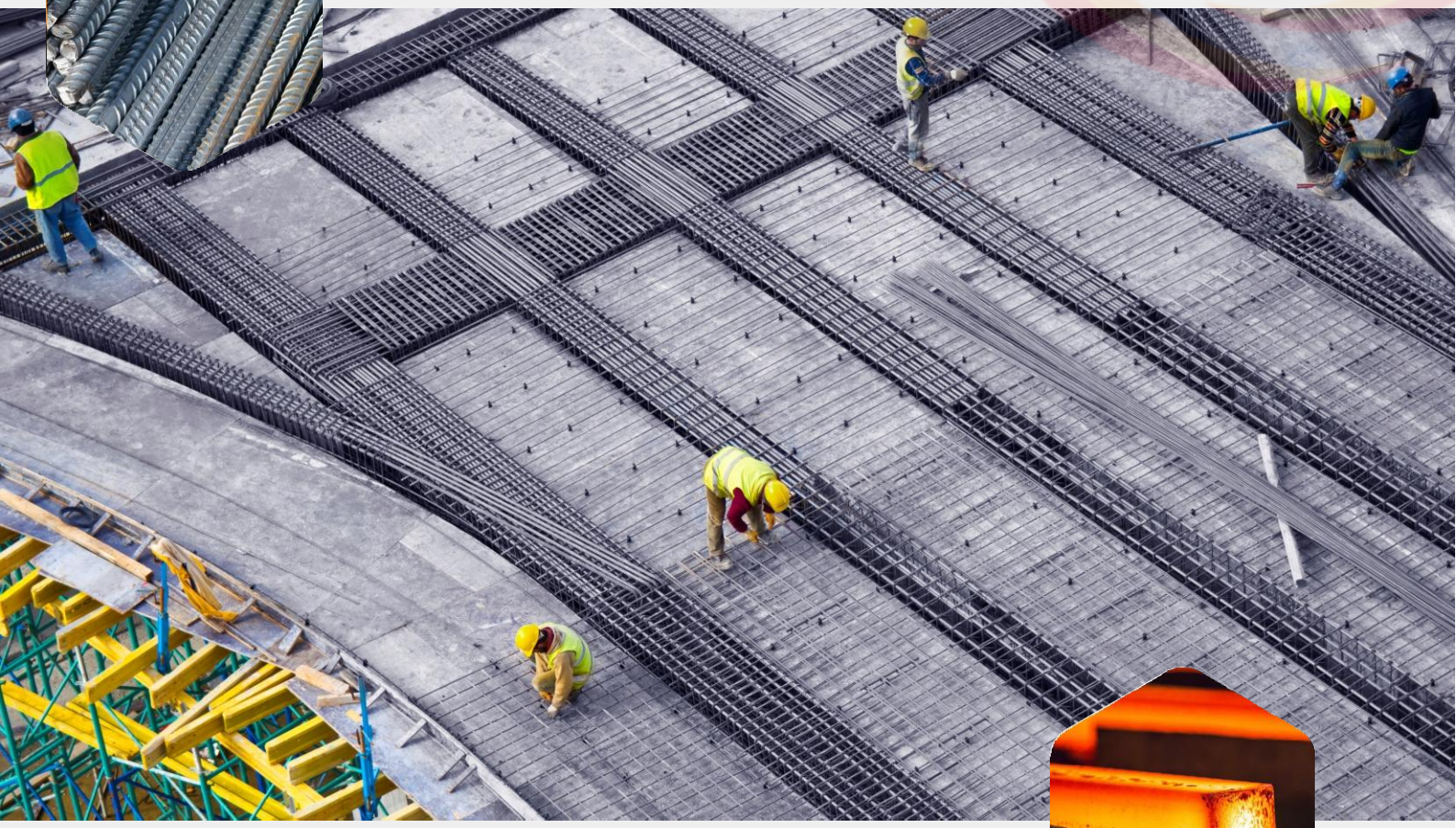
NUOR STEEL
حديد نور



BILLET PURCHASING SPECIFICATIONS SHEET FOR REBAR

➢ Specifications & Technical requirements of the Supplied Billet:

➢ Chemical parameters:

Chemistry	C	Si	Mn	P	S	Ni	Cr	Cu	Mn/S Ratio	Mn/Si Ratio
Percentage	0.20-0.26	0.20-0.25	0.60-0.70	0.040 max	0.040 max	0.20 max	0.20 max	0.25 max	20 min	3.0 min
Carbon Equivalent Value (CEV) ≤ 0.5										

1) Conformance to Standards: Any elements not explicitly specified herein shall conform to the requirements outlined in BS EN 10020.

   Note: In cases where the Carbon Equivalent Value (CEV) is calculated excluding incidental elements, adjustments to % Mn and %C shall be made as necessary to meet the required CEV.

2) Dimensions: Billets shall have dimensions of 130 mm x 130 mm with a length of 12000 mm.

3) Physical Properties: The physical properties shall align with the guidelines provided in Appendix

4) Material Test Certificate (MTC): Suppliers are required to furnish MTCs that include details of the chemical composition (minimum, maximum, and actual values) as well as the Carbon Equivalent (CEQ). The MTC should also specify the quantity of billets produced in each cast.

5) Marking and Identification: Each billet must be clearly marked with the supplier's name, heat number, and billet grade at both ends. Additionally, color-coded paint shall be applied to both ends.

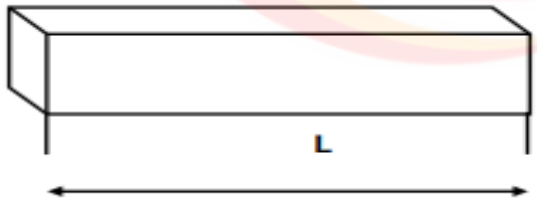
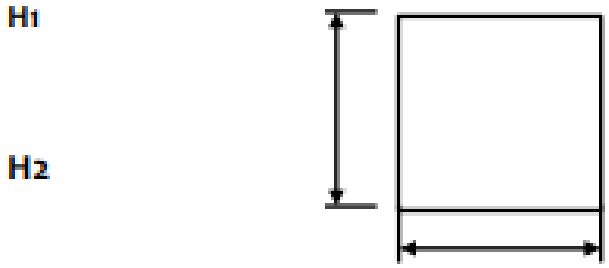
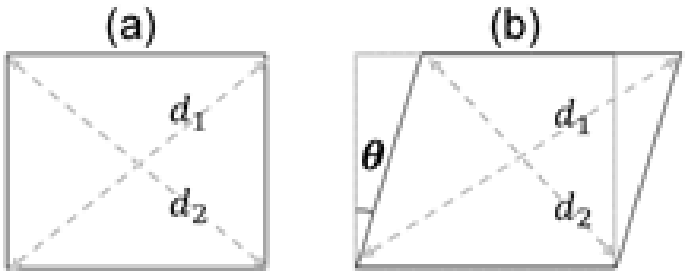
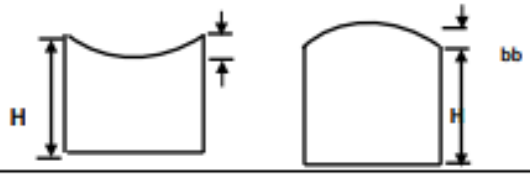
6) Information in MTC: The MTC must provide information regarding the process route, internal integrity, supply condition, customer's name, and the specified grade (B500B).

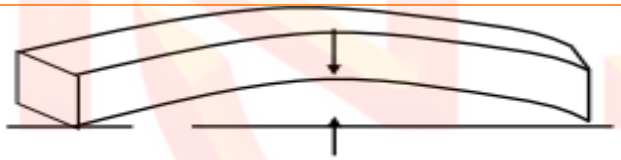
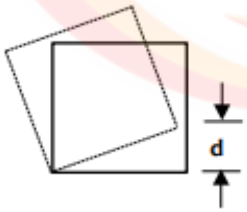
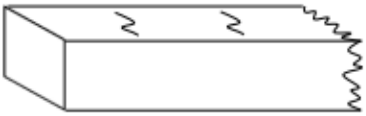
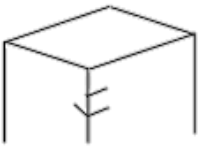
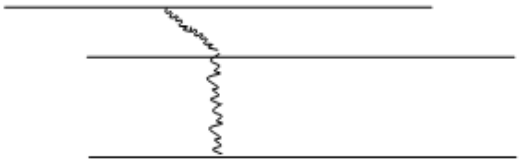
7) End Cutting: Billets may undergo end cutting through either hot saw or gas cutting methods. Hot shearing is acceptable, provided the billets maintain their square shape and both ends remain unbent. Billets failing to meet these criteria will be rejected.

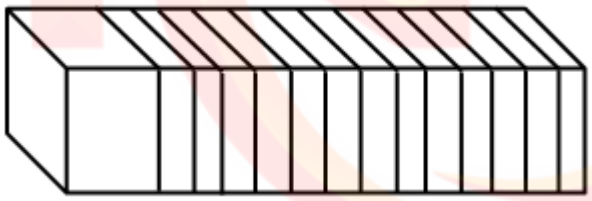
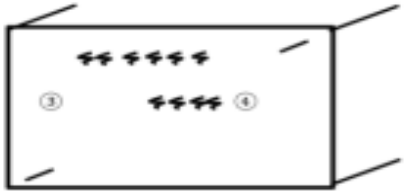
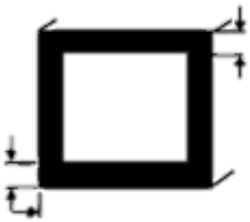
8) Radioactive Elements: Billets must be devoid of any radioactive substances. This requirement shall be reflected in the test certificate.

9) Verification of Chemical Analysis: Prior to billet shipment, PM4 shall receive a copy of the chemical analysis for billet heats, ensuring confirmation of compliance.

➢ Appendix 1: Physical Properties Reference

SI #	Description	Acceptance Limit (Max)	Schematic of defect
1.	**Length**	**5 % Max**	
2.	**Sections**	**H1 ± 2 mm** **H2 ± 2 mm**	
3.	**Rhomboidity**	**5% MAX** $\frac{d1 - d2 \text{ x } 100}{[0.5\, x\, (d1 + \text{d2})]}$	
4.	**Concavity / Convexity**	**Max 2 % Nominal Section size** $\left(\frac{b}{h}\right) x100 \leq 2\ \%$	

5.	**Straightness**	**5 mm / Meter. Max 60 mm in full length**	
6.	**Angular Twist**	**Max 5 degree In 12 meter**	
7.	**Longitudinal Cracks**	**Without any crack**	
8.	**Transverse Cracks**	**Without any crack**	
9.	**Corner Cracks**	**Without any crack**	
10.	**Casting arrest**	**Without any casting arrest**	

11.	**Oscillation Marks**	**Less than 2 mm depth**	
12.	**Crack in Section**	**Without any crack**	
13.	**Pinholes & Blowholes**	**Pinhole less than 2mm depth & without any blow hole or piping**	
14.	**Shrinkage Porosity**	**Without any Shrinkage Porosity**